

# App Dev Project Report

## 1. Student Details

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## 2. Project Details

**Project Title:** Trekking Management app

### **Problem Statement:**

To build a trekking management application that allows admin, trekker, staff to interact with application based on their role.

### **Approach:**

The app was built using Flask as the backend framework SQLAlchemy ORM for database management , HTML/CSS/BOOTSTRAP and jinja 2 template for frontend and SQLite as the database

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## 3. AI/LLM Declaration

I used LLM to assist in this project . The extent of LLM usage is around 10 -15%.

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## 4. Technologies and Frameworks Used

| Technology / Library | Purpose                                          |
|----------------------|--------------------------------------------------|
| Flask                | Core backend web framework                       |
| SQLAlchemy           | Object Relational Mapper for SQLite database     |
| Jinja2               | Template engine for rendering dynamic HTML pages |
| Bootstrap 5          | Frontend styling and responsive design           |
| Flask-Login          | User authentication and session management       |
| SQLite               | Lightweight local database for storing user data |

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## 5. Database Schema / ER Diagram

### Tables:

**User** — stores user profile details (id, name, email, password)

**Trek** — logs user trekking (id, staff\_id, duration, date,status)

**Booking**— (id, user\_id, staff\_id)

### Relationships:

One-to-Many → User → Booking

One user can have many bookings

One-to-Many → User → Trek

One staff have many assigned trek

## 6.Video Presentation

Drive Link:

<https://drive.google.com/file/d/1WTDnIV4Ze74dCwfPPjFF6Y6TT1EXxp1d/view?usp=sharing>